ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. ECE ENGINEERING â€" SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) Define Discrete Fourier Transform (DFT). Explain circular convolution and linear convolution using DFT. [7 Marks]
(b) Compute 8-point DFT of sequence x(n) = {1, 2, 2, 1, 1, 2, 2, 1} using Radix-2 Decimation-in-Time (DIT) FFT algorithm. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Derive butterfly signal flow graph for Radix-2 Decimation-in-Frequency (DIF) FFT algorithm. [7 Marks]
(b) Explain circular shift property, symmetry property, and Parseval's theorem of DFT. [7 Marks]

--- UNIT II ---

Q.2 (a) Compare characteristics of IIR and FIR digital filters. [7 Marks]
(b) Design an analog Butterworth low-pass filter with passband attenuation 1 dB at 1000 Hz and stopband attenuation 30 dB at 3000 Hz. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Explain Bilinear Transformation method of designing IIR digital filters. What is Frequency Warping? [7 Marks]
(b) Design a digital IIR filter using Impulse Invariance method from an analog transfer function H(s). [7 Marks]

--- UNIT III ---

Q.3 (a) Discuss design of linear phase FIR filters using Rectangular, Hamming, and Hanning window functions. [7 Marks]
(b) Design a high-pass FIR filter with cut-off frequency 0.5 rad/s using Hamming window of length M = 7. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Explain Frequency Sampling method for designing FIR filters with linear phase equations. [7 Marks]
(b) Discuss the Gibbs Phenomenon in FIR filter truncation. [7 Marks]

--- UNIT IV ---

Q.4 (a) Realize digital filter transfer function H(z) using Direct Form-I, Direct Form-II, Cascade, and Parallel structures. [7 Marks]
(b) Explain finite word-length effects in digital filters: coefficient quantization, roundoff noise, and limit cycles. [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Derive expression for signal-to-quantization noise ratio (SQNR) in A/D conversion. [7 Marks]
(b) Explain Lattice-Ladder realization of IIR and FIR filters. [7 Marks]

--- UNIT V ---

Q.5 (a) Describe multirate digital signal processing: Decimation and Interpolation by integer factor D and I with spectra. [7 Marks]
(b) Explain architecture of TMS320C67xx digital signal processor: Harvard architecture, pipelining, and MAC unit. [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Describe applications of DSP in speech processing and radar echo detection. [7 Marks]
(b) Explain Sub-band coding and Polyphase filter bank structures. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****